

TAPCO: LLM-Guided Design-Adaptive Timing and Power Co-Optimization

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Jeju Island, Republic of Korea | September 8, 2026

doi.org/10.1145/3831599.3840364

Limits of a fixed repair policy

MOTIVATION

The hard cases are not always missing a better number — sometimes the optimizer is making the wrong decision.

Fixed interface

Complex priors and failure patterns may not map to exposed controls.

Coupled controls

Margins, budgets, moves, and utilization interact by design.

Policy limitation

Corner cases need new schedules or commit logic, not new values.

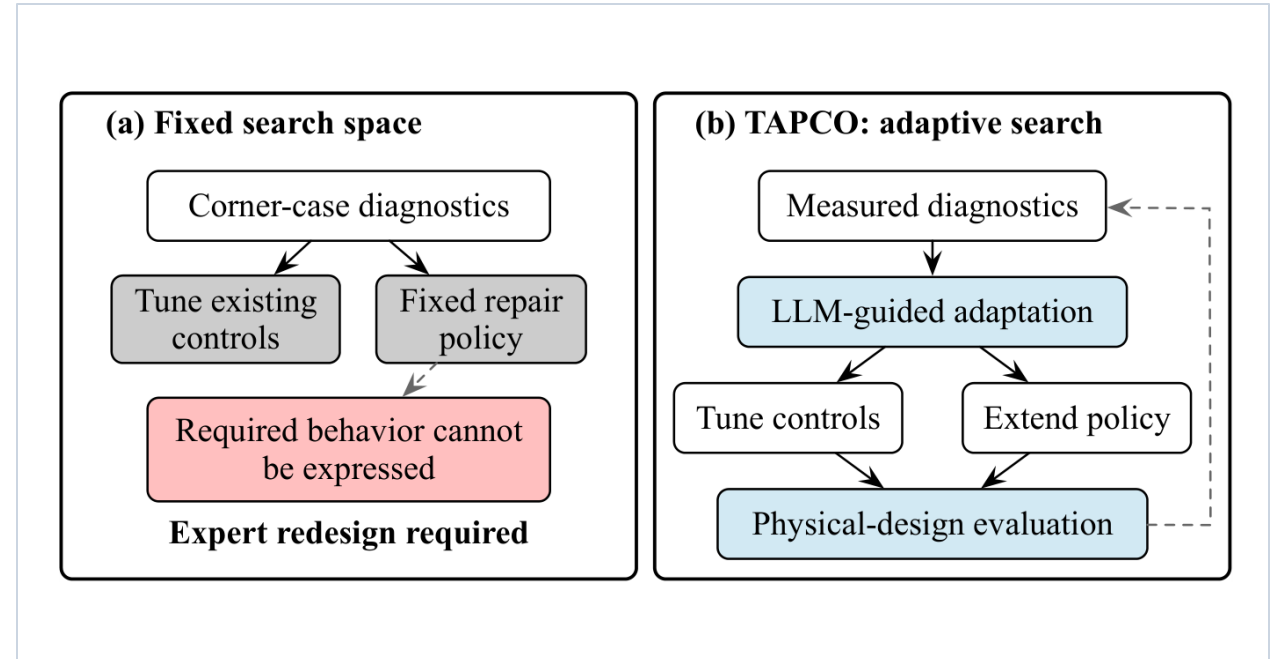


Fig. 1 Fixed versus adaptive optimizer search.

Retain efficient parameter tuning, but extend policy when measured behavior exposes a gap.

A candidate optimizer is the pair $\pi = (C, \theta)$: a source policy and its exposed controls.

SOURCE POLICY C · a policy edit δC opens a new slice

Endpoint schedule Effort allocation Partial-gain retention Stagnation recovery Candidate order

PARAMETER POLICY θ · search inside the slice $\Omega d(C)$

Timing / electrical margins Resource budgets Utilization caps Coverage Move switches

PHYSICAL-DESIGN KERNEL · we do not change this layer

Sizing VT substitution Buffering Cloning Pin swapping

Tune θ → diagnose a gap → edit C → re-tune θ in the new slice

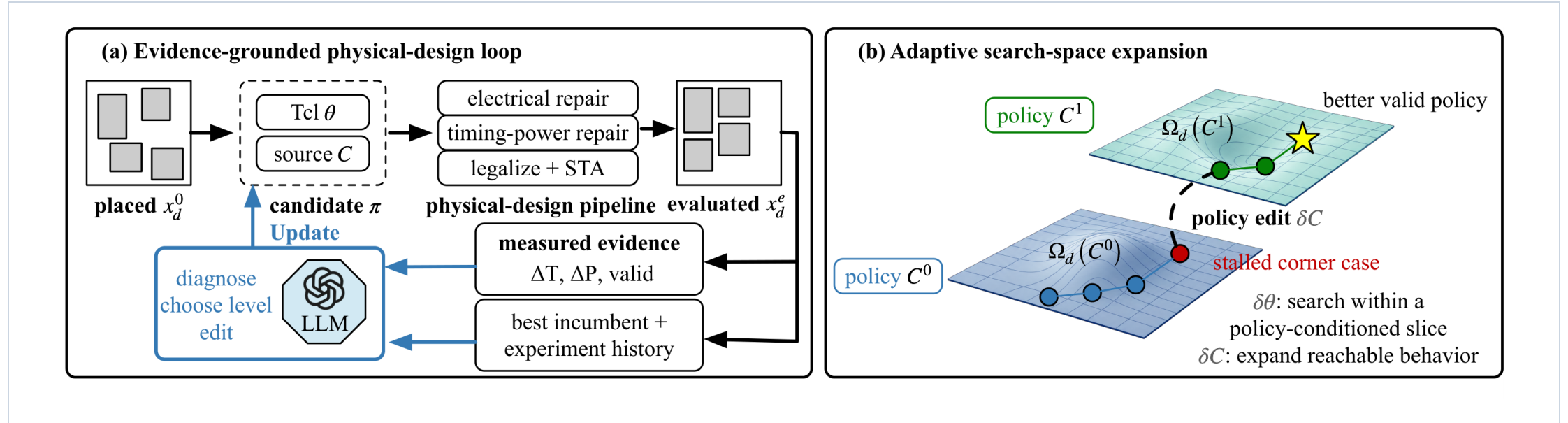


Fig. 2 (a) Evidence-grounded loop. (b) Parameter edits stay in $\Omega_d(C)$; a policy edit δC opens a new slice.

Each iteration: **diagnose** → **choose level** → **one bounded edit** → **replay from x^0** → **update evidence**

LLM AUTHORITY

Diagnose the bottleneck and choose one informative edit.

EVALUATOR AUTHORITY

Decide feasibility, objective value, and whether the incumbent changes.

Evidence, not a fixed round count, triggers a policy edit.

SYMPTOMS WE WATCH FOR

- 1. Endpoints stay unrepaired as effort rises
- 2. Work goes to shallow violations while deep paths dominate
- 3. Partial gains discarded by an all-or-nothing commit rule

Capability	Parameter-space limit	Source-level policy
Dynamic priority	Cannot update endpoint order as timing changes.	Recompute and re-sort the violating frontier.
Adaptive effort	Global budgets ignore violation depth and prior effort.	Allocate by severity; avoid repeated heavy repair.
Retention/recovery	Cannot change commit or stagnation behavior.	Retain partial gains; permit bounded recovery.
Candidate set	Move switches cannot define or order candidates.	Add VT/pin-swap pre-passes; select the equivalent-cell policy.

Table 3 Source-level adaptation beyond fixed parameters.

A larger budget repeats the same decision more aggressively — it does not change it.

EVALUATOR OBJECTIVE · Eq. (4) with $w_T, w_D, w_L = 30, 50, 50$

$$S = 30 \cdot \Delta T + 50 \cdot \Delta P_{\text{dyn}} + 50 \cdot \Delta P_{\text{leak}} - A_{\text{eval}}$$

A_{eval} collects electrical, runtime, displacement, and routing penalties. Legality and equivalence must both hold.

Benchmark	Physical-design flow	Baseline	LLM
Eight ASAP7 designs: AES, JPEG, Ariane, NVDLA-P / C / A / M, and Mempool.	OpenROAD Resizer, evaluated with OpenROAD and OpenSTA.	The contest reference flow, on the same placed designs.	Optimizer development used OpenAI Codex 5.5.

REPORTING BOUNDARY

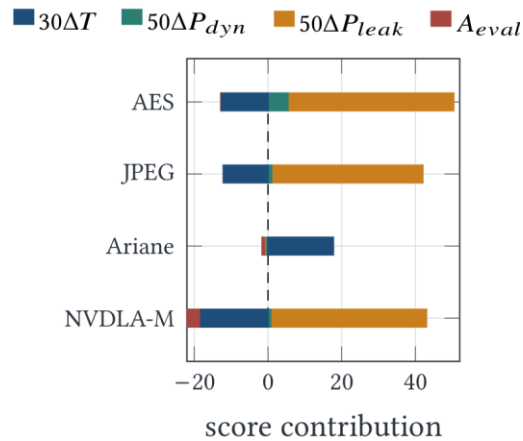
Raw final-phase QoR is reported for the four valid designs. NVDLA-P and NVDLA-C fail the flip-flop integrity check; NVDLA-A and Mempool are earlier-phase results.

Results: QoR and score composition

RESULTS

Design	Baseline			TAPCO			
	TNS (ns)	P_{dyn}	P_{leak}	TNS (ns)	P_{dyn}	P_{leak}	Score
AES	−12.44	431.904	0.0963	−17.67	384.990	0.0097	37.5818
JPEG	−48.77	293.826	0.1740	−68.51	287.969	0.0315	29.8166
Ariane	−7568.43	640.100	17.900	−3092.71	645.000	18.000	16.1389
NVDLA-M	−13.99	44.539	0.0615	−22.49	43.891	0.0095	20.8050

Table 1 Raw QoR on the four valid final-phase designs. Power in mW; TNS negative as reported.



(a) Score decomposition

Official rank: 2nd of the MLCAD 2026 contest

Aggregate scores 40.3031 and 47.5586 under the two schemes.

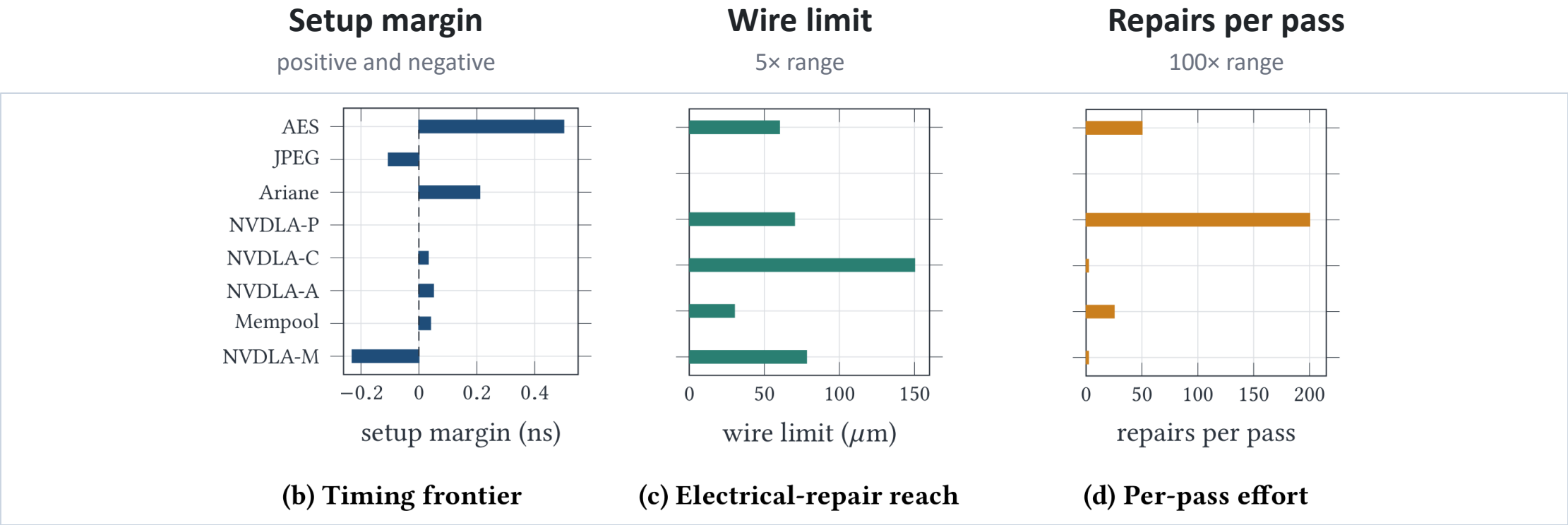
Leakage-oriented — AES, JPEG, NVDLA-M

Leakage gains of 45.0, 40.9 and 42.3 points beat TNS losses of 12.6, 12.1 and 18.2. AES leakage: 0.0963 → 0.0097 mW.

Timing-oriented — Ariane

17.7 timing points at under one point of power cost. TNS: −7568 → −3093 ns.

Across all eight optimizers the controls do not converge to one configuration.



TAPCO adapts the optimizer itself, not only its exposed parameters.

- | | | |
|---|------------------------------|--|
| 1 | Program prior | The LLM connects source code, diagnostics and strategies to propose one bounded edit. |
| 2 | Evaluator grounding | The physical-design pipeline owns validity, objective value and acceptance. |
| 3 | Adaptive search space | Tune controls first; extend the source policy only when the interface limits behavior. |

Independent evaluation ranked TAPCO 2nd overall and yielded distinct timing- and leakage-oriented policies.



One iteration, step by step

Each iteration tests one hypothesis with one bounded edit.

1	2	3	4	5
Diagnose	→ Choose level	→ Apply one edit	→ Replay from x^0	→ Update evidence
Score decomposition, endpoint slack, violations and failure logs.	A parameter edit if the interface is expressive; otherwise a source edit.	Bound the edited region so credit assignment stays clear.	Every candidate starts from the same placed design.	Accept only a valid improvement; append every attempt.

The retained incumbent never gets worse; failures remain useful evidence.

THREE WAYS A CANDIDATE CAN FAIL — ALL THREE ARE RECORDED

Does not build	Not valid	Valid but worse
The edit fails to compile, or the flow crashes.	Legality or equivalence fails, whatever the score.	The incumbent is kept; the attempt still informs the next edit.

Design-specific parameter settings

Default denotes the inherited setting of the common repair flow.

Design	Wire limit (μm)	Slew/cap. margin (%)	Setup margin (ns)	Repairs/pass
AES	60	10/10	0.500	50
JPEG	Default	Default	−0.105	Default
Ariane	70	Default	0.210	200
NVDLA-P	150	19/19	Default	2
NVDLA-C	30	20/20	0.032	25
NVDLA-A	78	9/9	0.050	2
Mempool	Default	Default	0.040	Default
NVDLA-M	Default	Default	−0.230	Default

Table 2 Key design-specific parameter settings.

Move sets differ too: AES restricts buffering, VT swapping and last-gasp repair; Ariane keeps the broad move set.

Why four designs in Table 1

The evaluator runs four validity checks:

- 1 Physical cell locations unchanged
- 2 Macro locations unchanged
- 3 I/O port locations unchanged
- 4 **Flip-flop integrity**

NVDLA-P and NVDLA-C failed check 4. It compares the flip-flop count, and each flip-flop's clock net with the cell driving it.

So it is stricter than logic equivalence: a legal, identical netlist still fails if a clock driver was resized.

Both passed in earlier phases — this is a final-phase regression.

The adjustment is almost entirely runtime

Design	S_PPA	Final	Δ	Runtime
NVDLA-M	24.78	20.81	-3.98	5.08×
Ariane	17.08	16.14	-0.94	1.75×
AES	37.80	37.58	-0.22	1.20×
JPEG	29.80	29.82	+0.01	0.88×

The gap between S_PPA and the final score is the runtime penalty, not a QoR effect. NVDLA-M took 233 s against a 46 s baseline, costing 3.93 points alone.

Per-phase score history

NVDLA-P and NVDLA-C were valid until the final phase.

Design	Beta 1	Beta 2	Beta 3 (final)	Final-phase status
AES	37.32	37.58	37.58	valid
JPEG	29.81	29.72	29.82	valid
Ariane	16.16	16.16	16.14	valid
NVDLA-M	—	—	20.81	valid
NVDLA-P	31.14	43.19	—	failed check 4
NVDLA-C	23.77	25.20	—	failed check 4
NVDLA-A	21.49	—	—	earlier-phase result
Mempool	—	7.13	—	earlier-phase result

NVDLA-P scored 43.19 in beta 2 — our best single-design result in any phase — and then regressed to invalid.

Optimizer adaptation is an expensive black-box search: a program change must also build, and can fail on functionality, legality, or runtime.

	Auto-tuner / Bayesian optimization	TAPCO
What is searched	Exposed margins, budgets, move switches and utilization limits.	The same controls, plus the source policy that decides how they are used.
Assumption	The fixed policy can already express the needed behavior.	None. The policy itself is part of the search space.
When it stalls	A larger budget repeats the same decision more aggressively.	Diagnose the gap, edit the decision, then resume parameter tuning.

The two levels work together: parameters specialize the envelope; source policies change how it is consumed.

The artifact freezes the per-design optimizers and verifies the paper's claims without rerunning the LLM search.

Included

Final Tcl and source snapshots, organizer results, benchmark inputs, validation scripts.

Runtime

Ubuntu 24.04 Docker image; 32 GB RAM and 80 GB disk recommended.

No external dependency

No GPU, commercial EDA license, or LLM API call.

What verify reproduces

The four valid results, the score decomposition, and the policy settings.

Evaluation commands

```
bash run.sh smoke
```

```
bash run.sh verify
```

```
bash run.sh full (optional)
```



ARTIFACT

[10.5281/zenodo.21566683](https://doi.org/10.5281/zenodo.21566683)

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